



NEET TEST HUB

NTH-PREMIUM TEST SERIES 15

Date – 18 January 2026

Duration – 3 Hours

Marks – 720

Physics :- Electric Charges & Fields, Electrostatic Potential & Capacitance, Current Electricity, Moving Charges & Magnetism, Magnetism & Matter, EMI, Alternating Current, Electromagnetic Waves

Chemistry :- Solutions, Electrochemistry, Chemical Kinetics, The d & f-Block Elements, Coordination Compounds, The p-Block Elements (Group 13 to 18), Salt Analysis

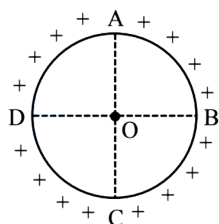
Biology :- Sexual Reproduction in Flowering Plants, Human Reproduction , Reproductive Health, Principles of Inheritance & Variation, Molecular Basis of Inheritance, Evolution

Instruction

1. The test is of 3 hours duration.
2. The test booklet consists of 200 questions. The maximum mark is 720.
3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,
4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc AB to 'O' is 'E' its magnitude. What would be the magnitude of electric field at 'O' due to arc ABC?



- A) $2E$ B) $\sqrt{2}E$ C) $E/2$ D) Zero
2. The magnetic field in a plane electromagnetic wave is given by $B = 2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$. This electromagnetic wave is

- A) Visible light B) Infrared
C) Microwave D) Radiowave
3. Match List I with List II. Choose the correct answer from the options given below

List I		List II	
A.	$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$	I.	Gauss' law for electricity
B.	$\oint \vec{E} \cdot d\vec{l} = \frac{d\phi_B}{dt}$	II.	Gauss' law for magnetism
C.	$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$	III.	Faraday law
D.	$\oint \vec{B} \cdot d\vec{A} = 0$	IV.	Ampere - Maxwell law

- A) A - IV, B - I, C - III, D - II
B) A - II, B - III, C - I, D - IV
C) A - IV, B - III, C - I, D - II
D) A - I, B - II, C - III, D - IV

4. Identify the correct statements from the following descriptions of various properties of electromagnetic waves.

A. In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.

B. The energy in electromagnetic wave is divided equally between electric and magnetic fields.

C. Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.

D. The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.

E. The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.

Choose the most appropriate answer from the options given below:

- A) D only
B) B and D only
C) B, C and E only
D) A, B and E only

5. A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is.... Vm^{-1}

- A) 2.45 B) 5.48
C) 7.75 D) 9.73

6. In which of the following, emission of electrons does not take place

- A) thermionic emission
B) X-rays emission
C) photoelectric emission
D) secondary emission

7. The instantaneous voltages at three terminals marked X, Y and Z are given by

$$V_x = V_0 \sin \omega t \quad V_y = V_0 \sin \left(\omega t + \frac{2\pi}{3} \right) \quad \text{and} \quad V_z = V_0 \sin \left(\omega t + \frac{4\pi}{3} \right)$$

An ideal voltmeter is configured to read rms value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be

$$[A] V_{XY}^{rms} = V_0 \sqrt{\frac{3}{2}}$$

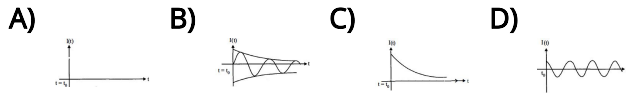
$$[B] V_{YZ}^{rms} = V_0 \sqrt{\frac{1}{2}}$$

$$[C] V_{XY}^{rms} = V_0$$

[D] independent of the choice of the two terminals

A) A, C B) A, B C) A, D D) A, C, D

8. A series LR circuit is connected to a voltage source with $V(t) = V_0 \sin \omega t$. After very large time, current $I(t)$ behaves as ($t_0 \gg \frac{L}{R}$)



9. If an AC main supply is given to be $220 V$. The average emf during a positive half cycle will be..... V

A) 198 B) 200 C) 240 D) $200\sqrt{2}$

10. The root mean square value of the alternating current is equal to

A) Twice the peak value
B) Half the peak value
C) $\frac{1}{\sqrt{2}}$ times the peak value
D) Equal to the peak value

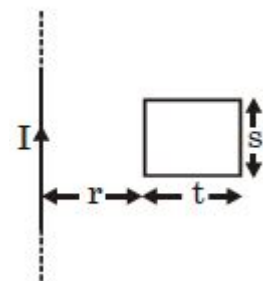
11. The r.m.s. voltage of domestic electricity supply is 220 volt . Electrical appliances should be designed to withstand an instantaneous voltage of..... V

A) 20 B) 310 C) 330 D) 440

12. The magnetic flux linked with a coil of N turns of area of cross section A held with its plane parallel to the field B is

A) $\frac{NAB}{2}$ B) NAB C) $\frac{NAB}{4}$ D) zero

13. Flux passing through loop is proportional to



A) $\phi \propto r$
B) $\phi \propto s$
C) $\phi \propto t$
D) $\phi \propto r^{-1}$

14. A closed planar wire loop of area A and arbitrary shape is placed in a uniform magnetic field of magnitude B , with its plane perpendicular to magnetic field. The resistance of the wire loop is R . The loop is now turned upside down by 180° so that its plane again becomes perpendicular to the magnetic field. The total charge that must have flowed through the wire ring in the process is

A) $< AB/R$ B) $= AB/R$
C) $= 2AB/R$ D) None

15. A conducting rod of length l is falling with a velocity v perpendicular to a uniform horizontal magnetic field B . The potential difference between its two ends will be

A) $2Blv$ B) Blv
C) $\frac{1}{2}Blv$ D) $B^2 l^2 v^2$

16. If the magnetic flux is expressed in weber, then magnetic induction can be expressed in

A) Weber/m^2 B) Weber/m
C) $\text{Weber} - m$ D) $\text{Weber} - m^2$

17. The magnetic potential due to a magnetic dipole at a point on its axis situated at a distance of 20 cm from its center is $1.5 \times 10^{-5} \text{ Tm}$. The magnetic moment of the dipole is _____ Am^2 . (Given : $\frac{\mu_0}{4\pi} = 10^{-7} \text{ TmA}^{-1}$)

A) 6 B) 5 C) 4 D) 12

18. The unit of magnetic moment is

A) Wb/m B) $\text{Wb}.m^2$
C) $A.m$ D) $A.m^2$

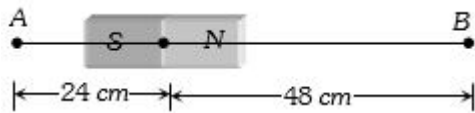
19. The magnetic moment of a magnet of length 10 cm and pole strength 4.0 Am will be..... Am^2

A) 0.4 B) 1.6 C) 20 D) 8

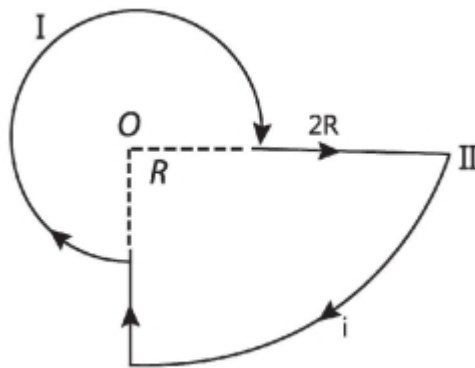
20. The magnetic compass is not useful for navigation near the magnetic poles because

A) The magnetic field near the poles is zero
B) The magnetic field near the poles is almost vertical
C) At low temperature, the compass needle loses its magnetic properties
D) Neither of the above

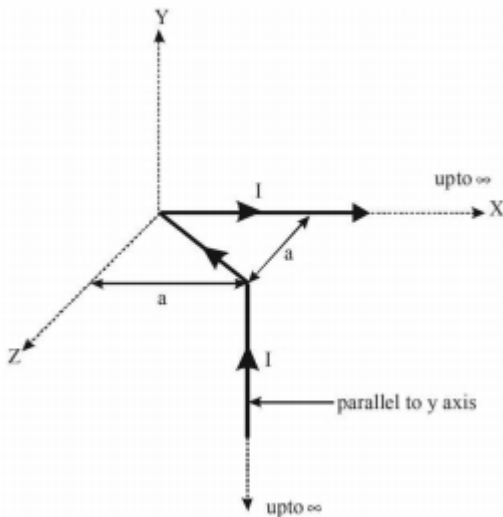
21. A bar magnet of length 3 cm has points A and B along its axis at distances of 24 cm and 48 cm on the opposite sides. Ratio of magnetic fields at these points will be



- A) 8 B) $1/2\sqrt{2}$ C) 3 D) 4
22. Consider the circular loop having current i and with central point O . The magnetic field at the central point O is



- A) $\frac{2\mu_0 i}{3\pi R}$ acting downward
 B) $\frac{5\mu_0 i}{12R}$ acting downward
 C) $\frac{6\mu_0 i}{11R}$ acting downward
 D) $\frac{3\mu_0 i}{7R}$ acting upward
23. The magnetic field at the origin due to the current flowing in the wire is -



- A) $-\frac{\mu_0 I}{8\pi a}(\hat{i} + \hat{k})$
 B) $\frac{\mu_0 I}{2\pi a}(\hat{i} + \hat{k})$
 C) $\frac{\mu_0 I}{8\pi a}(-\hat{i} + \hat{k})$
 D) $\frac{\mu_0 I}{4\pi a\sqrt{2}}(\hat{i} - \hat{k})$
24. A particle is moving with velocity $\vec{v} = \hat{i} + 3\hat{j}$ and it produces an electric field at a point given by $\vec{E} = 2\hat{k}$. It will produce magnetic field at that point equal to (all quantities are in SI units)

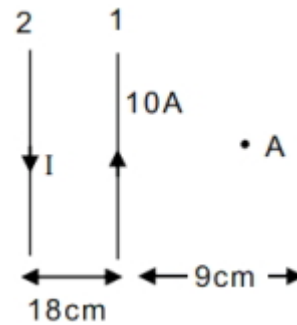
A) $\frac{6\hat{i} - 2\hat{j}}{c^2}$

B) $\frac{6\hat{i} + 2\hat{j}}{c^2}$

C) zero

D) cannot be determined from the given data

25. In figure two parallel infinitely long current carrying wires are shown. If resultant magnetic field at point A is zero. Then determine current I . (in A)

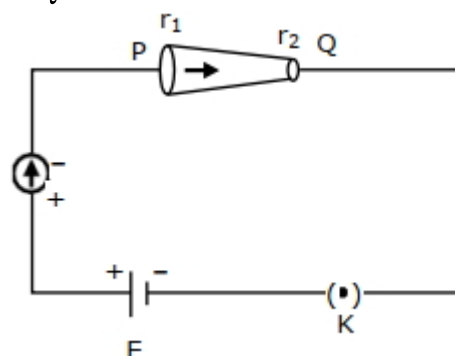


- A) 50 B) 15 C) 30 D) 25
26. An electron (charge q coulomb) enters a magnetic field of H weber/ m^2 with a velocity of v m/s in the same direction as that of the field the force on the electron is
- A) Hqv Newton's in the direction of the magnetic field
 B) Hqv dynes in the direction of the magnetic field
 C) Hqv Newton's at right angles to the direction of the magnetic field
 D) Zero
27. A bird sitting on a single high tension wire does not get electrocuted because

- A) the circuit is not complete
 B) resistance of the bird is too high
 C) capacitance of the bird is too small and the line frequency is too small
 D) Both a and b

28. In the given figure, a battery of emf E is connected across a conductor PQ of length ' Y ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$).

Choose the correct option as one moves from P to Q :



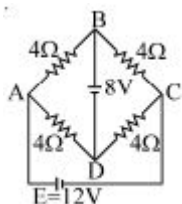
- A) Electric field decreases.

- B) Drift velocity of electron increases.
 C) Electron current decreases.
 D) All of these

29. When 5 V potential difference is applied across a wire of length 0.1 m , the drift speed of electrons is $2.5 \times 10^{-4}\text{ ms}^{-1}$. If the electron density in the wire is $8 \times 10^{28}\text{ m}^{-3}$ the resistivity of the material is close to

- A) $1.6 \times 10^{-7}\text{ }\Omega\text{m}$ B) $1.6 \times 10^{-5}\text{ }\Omega\text{m}$
 C) $1.6 \times 10^{-3}\text{ }\Omega\text{m}$ D) $1.6 \times 10^{-8}\text{ }\Omega\text{m}$

30. The given figure shows a network of resistances and a battery. Which of the two batteries is getting charged?

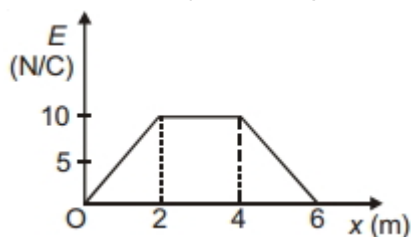


- A) 8 V battery B) 12 V battery
 C) none D) can't be said

31. A wire of diameter 0.02 meter contains 10^{28} free electrons per cubic meter. For an electrical current of 100 A , the drift velocity of the free electrons in the wire is nearly

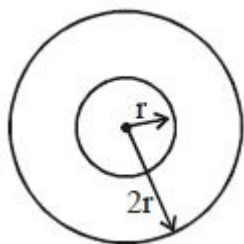
- A) $1 \times 10^{-19}\text{ m/s}$ B) $5 \times 10^{-10}\text{ m/s}$
 C) $2 \times 10^{-4}\text{ m/s}$ D) $8 \times 10^3\text{ m/s}$

32. Figure shows the variation of electric field intensity E versus distance x . What is the potential difference between the points at $x = 2\text{ m}$ and at $x = 6\text{ m}$ from O is V



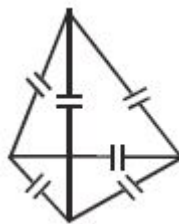
- A) 30 B) 60 C) 40 D) 80

33. If the potential of the inner shell is 10 V and that of the outer shell is 5 V , then potential at the centre will be.... V



- A) 10 B) 5 C) 15 D) 0

34. If the capacitance of each capacitor in the tetrahedron shown is C , then effective capacitance of the network across any two junctions is C



- A) 2 B) 1 C) 0.5 D) 5

35. In a region, if electric field is defined as $\vec{E} = (\hat{i} + 2\hat{j} + \hat{k})\text{ V/m}$, then the potential difference between two points $A(0,0,0)$ and $B(2,3,4)$ in that region, is V

- A) 6 B) 12 C) 8 D) 9

36. Equal charges are given to two spheres of different radii. The potential will

- A) Be more on the smaller sphere
 B) Be more on the bigger sphere
 C) Be equal on both the spheres
 D) Depend on the nature of the materials of the spheres

37. An infinite plane sheet of charge having uniform surface charge density $+\sigma_5\text{ C/m}^2$ is placed on $x-y$ plane. Another infinitely long line charge having uniform linear charge density $+\lambda_e\text{ C/m}$ is placed at $z = 4\text{ m}$ plane and parallel to y -axis. If the magnitude values $|\sigma_s| = 2|\lambda_e|$ then at point $(0,0,2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$. The value of n is _____.

- A) 16 B) 20 C) 23 D) 30

38. Find out the surface charge density at the intersection of point $x = 3\text{ m}$ plane and x -axis, in the region of uniform line charge of 8 nC/m lying along the z -axis in free space.

- A) 0.424 nCm^{-2}
 B) 47.88 C/m
 C) 0.07 nCm^{-2}
 D) 4.0 nCm^{-2}

39. Select the correct statement about electric charge

- A) Charge can be converted into energy and energy can be converted into charge
 B) Charge of a particle increases with increase in its velocity

- C) Charge on a body is always integral multiple of a certain charge called charge of electron
D) Charge on a body is always positive or zero

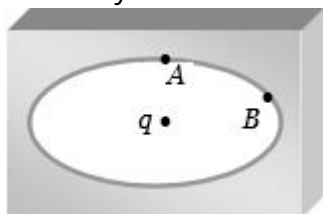
40. Two identical tennis balls each having mass m and charge q are suspended from a fixed point by threads of length l . What is the equilibrium separation when each thread makes a small angle θ with the vertical?

- A) $x = \left(\frac{q^2 l}{2\pi\epsilon_0 m g} \right)^{1/2}$
B) $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g^2} \right)^{1/3}$
C) $x = \left(\frac{q^2 l}{2\pi\epsilon_0 m g} \right)^{1/3}$
D) $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g} \right)^{1/3}$

41. A neutral water molecule (H_2O) in its vapor state has an electric dipole moment of magnitude $6.4 \times 10^{-30} C - m$. How far apart are the molecules centres of positive and negative charge

- A) $4 m$ B) $4 mm$
C) $4 pm$ D) $4 \mu m$

42. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the centre of the cavity. The points A and B are on the cavity surface as shown in the figure. Then



- A) Electric field near A in the cavity = Electric field near B in the cavity
B) Total electric field flux through the surface of the cavity is q/ϵ_0
C) Potential at A = Potential at B
D) Both (b) and (c)

43. Three charges each of magnitude q are placed at the corners of an equilateral triangle, the electrostatic force on the charge placed at the center is (each side of triangle is L)

- A) Zero
B) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{L^2}$
C) $\frac{1}{4\pi\epsilon_0} \frac{3q^2}{L^2}$
D) $\frac{1}{12\pi\epsilon_0} \frac{q^2}{L^2}$

44. One metallic sphere A is given positive charge whereas another identical metallic sphere B of exactly same mass as of A is given equal amount of negative charge. Then

- A) Mass of A and mass of B still remain equal

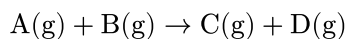
- B) Mass of A increases
C) Mass of B decreases
D) Mass of B increases

45. Charge on α - particle is

- A) $4.8 \times 10^{-19} C$ B) $1.6 \times 10^{-19} C$
C) $3.2 \times 10^{-19} C$ D) $6.4 \times 10^{-19} C$

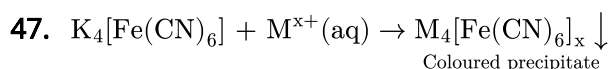
Chemistry

46. Consider an elementary reaction



If the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become ' x ' times of the original reaction rate. The value of x is :

- A) 3 B) $\frac{1}{9}$ C) 9 D) $\frac{1}{3}$



Which of the following cation does not respond to the above reaction?

- A) $Cu^{2+}(aq)$ B) $Fe^{3+}(aq)$
C) $Zn^{2+}(aq)$ D) None of these

48. Calculate the number of unpaired electrons in $[Mn(H_2O)_6]^{2+}$, Considering H_2O as a weak field ligand (At. No. of $Mn = 25$)

- A) Two B) Three C) Four D) Five

49. In which of the following octahedral complexes of Co (At. No. 27), will the magnitude of Δ_0 be the highest?

- A) $[Co(CN)_6]^{3-}$ B) $[Co(C_2O_4)_3]^{3-}$
C) $[Co(H_2O)_6]^{3+}$ D) $[Co(NH_3)_6]^{3+}$

50. $[Fe(NO_2)_3Cl_3]$ and $[Fe(O - NO)_3Cl_3]$ are

- A) linkage isomers
B) geometrical isomers
C) optical isomers
D) hydrate isomers

51. For the complex with molecular formula $CrCl_3 \cdot 6H_2O$ 0.1, mole of the complex when treated with excess of $AgNO_3$ gave 28.7 g of white precipitate of $AgCl$ (Molar mass = 143.5 g). The formula of the complex would be

- A) $[Cr(H_2O)_6]Cl_3$ B) $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$
C) $[CrCl_2(H_2O)_4]Cl \cdot 2H_2O$ D) $[Cr(H_2O)_3Cl_3] \cdot 3H_2O$

52. Potassium dichromate is used -

- A) As a reducing agent
B) In electroplating
C) As an insecticide

- D) as an oxidizing agent it oxidises ferrous ions into ferric ions in acidic media
53. The radius of La^{3+} (Atomic number of La=57) is 1.06 Å. Which one of the following given values will be closest to the radius of Lu^{3+} (Atomic number of Lu=71)?
 A) 1.60 Å B) 1.40 Å C) 1.06 Å D) 0.85 Å
54. Calculate the value of magnetic moment for $^{26}\text{Fe}^{3+}$.
 A) 1.73 BM B) 3.87 BM
 C) 4.90 BM D) 5.92 BM
55. The lanthanide contraction is responsible for the fact that
 A) Zr and Hf have same radius
 B) Zr and Zn have the same oxidation state
 C) Zr and Y have same radius
 D) Zr and Nb have similar oxidation state
56. Although Zirconium belongs to 4d and hafnium belongs to 5d transition series, they show similar physical and chemical properties because both
 A) belong to d-block
 B) have same number of electrons
 C) belongs to the same group of the periodic table
 D) have similar atomic radius
57. The coordination number of Cu in complex $[\text{Cu}(\text{H}_2\text{O})_4]^{++}$ is
 A) 4 B) 3 C) 2 D) 1
58. Co-ordination number of Fe in the complexes $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeCl}_4]^-$ would be respectively
 A) 2, 3, 3 B) 6, 6, 4
 C) 6, 3, 3 D) 6, 4, 6
59. A co-ordination complex compound of cobalt has the molecular formulae containing five ammonia molecules, one nitro group and two chlorine atoms for one cobalt atom. One mole of this compound produces three mole ions in an aqueous solution on reacting with excess of AgNO_3 , AgCl precipitate. The ionic formula for this complex would be:
 A) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$
 B) $[\text{Co}(\text{NH}_3)_5\text{Cl}][\text{Cl}(\text{NO}_2)]$
 C) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)\text{Cl}][(\text{NH}_3)\text{Cl}]$
 D) $[\text{Co}(\text{NH}_3)_5][(\text{NO}_2)_2\text{Cl}_2]$
60. Cobalt (III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test for chloride ions with silver nitrate at 25 °C?
 A) $\text{CoCl}_3 \cdot 5\text{NH}_3$ B) $\text{CoCl}_3 \cdot 6\text{NH}_3$
 C) $\text{CoCl}_3 \cdot 3\text{NH}_3$ D) $\text{CoCl}_3 \cdot 4\text{NH}_3$
61. Two isomers (A) and (B) with Molar mass 184 g/mol and elemental composition C, 52.2%; H, 4.9% and Br 42.9% gave benzoic acid and p-bromobenzoic acid, respectively on oxidation with KMnO_4 . Isomer 'A' is optically active and gives a pale yellow precipitate when warmed with alcoholic AgNO_3 . Isomer 'A' and 'B' are, respectively
 A)  B)  C)  D) 
62. Which d-block element is expected to have highest enthalpy of atomisation, among the following?
 A) Sc B) Mn
 C) V, Zn D) None of these
63. Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is incorrect?
 A) Ferrous oxide is more basic in nature than the ferric oxide.
 B) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.
 C) Ferrous compounds are less volatile than the corresponding ferric compounds.
 D) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.
64. The highest magnetic moment is shown by the transition metal ion with the outer electronic configuration
 A) $3d^2$ B) $3d^5$
 C) $3d^7$ D) $3d^9$
65. What is the name of element with atomic number 105
 A) Kurchatovium B) Dubnium C) Nobelium D) Holmium
66. Which of the following represents the electronic configuration of a transition element
 A) $1s^2, 2s^2p^6 \dots ns^2p^3$
 B) $1s^2, 2s^2p^6 \dots ns^2p^6d^3, (n+1)s^2$
 C) $1s^2, 2s^2p^6 \dots ns^2p^6d^{10}, (n+1)s^2p^1$
 D) $1s^2, 2s^2p^6 \dots ns^2p^6$

67. Rate constant varies with temperature by the equation $\log_{10} K = 5 - 2000/T$. We can conclude that ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
- A) Pre exponential factor A is 5
 B) E_a is 4 kcal/mol
 C) Pre exponential factor A is 10^5
 D) E_a is 19.212 kcal/mol
68. The rate of reaction $A + 2B \rightarrow 3C$ becomes 72 times when concentration of A is tripled and concentration of B is doubled then the order of reaction with respect to A and B respectively is
- A) 1, 2 B) 2, 3 C) 3, 2 D) 2, 2
69. NO_2 required for a reaction is produced by the decomposition of N_2O_5 in CCl_4 as per the equation $2\text{N}_2\text{O}_5(g) \rightarrow 4\text{NO}_2(g) + \text{O}_2(g)$ The initial concentration of N_2O_5 is 3.00 mol L^{-1} and it is 2.75 mol L^{-1} after 30 minutes. The rate of formation of NO_2 is $\text{mol L}^{-1} \text{ min}^{-1}$
- A) 1.667×10^{-2} B) 4.167×10^{-3}
 C) 8.333×10^{-3} D) 2.083×10^{-3}
70. In a first order reaction the $a/(a-x)$ was found to be 8 after 10 minute. The rate constant is
- A) $\frac{(2.303 \times 3 \log 2)}{10}$
 B) $\frac{(2.303 \times 2 \log 3)}{10}$
 C) $10 \times 2.303 \times 2 \log 3$
 D) $10 \times 2.303 \times 3 \log 2$
71. $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$. On increasing $[\text{Cu}^{2+}]$ concentration, electrode potential
- A) Increases
 B) Decreases
 C) Will remain unchanged
 D) First increases, then decreases
72. Calculate E_{cell}° of following galvanic cell at 298 K ; V
 $\text{Ca}(s) | \text{Ca}^{+2}(\text{aq}) || \text{Fe}^{+2}(\text{aq}) | \text{Fe}(s)$
 $E_{\text{Ca}^{+2}/\text{Ca}}^{\circ} = -2.87 \text{ V}$; $E_{\text{Fe}/\text{Fe}^{+2}}^{\circ} = 0.41 \text{ V}$
- A) 2.46 B) -2.46 C) 3.28 D) -3.28
73. Electrolysis of molten anhydrous calcium chloride produces
- A) Calcium B) Phosphorus C) Sulphur D) Sodium
74. Electrolysis rules of Faraday's states that mass deposited on electrode is proportional to
- A) $m \propto I^2$
 B) $m \propto Q$
 C) $m \propto Q^2$
 D) None of these
75. Electrolysis involves oxidation and reduction respectively at
- A) Anode and cathode
 B) Cathode and anode
 C) At both the electrodes
 D) None of the above
76. Mole fraction of O_2 in O_2 and O_3 mixture is 0.25. Hence percentage concentration ($\frac{w}{W}\%$) O_2 in mixture is
- A) 18.18 B) 1.81 C) 81.82 D) 8.18
77. On mixing, heptane and octane form an ideal solution. At 373 K , the vapour pressures of the two liquid components (heptane and octane) are 105 kPa and 45 kPa respectively. Vapour pressure of the solution obtained by mixing 25.0 g of heptane and 35 g of octane will be kPa . (molar mass of heptane = 100 g mol^{-1} and of octane = 114 g mol^{-1})
- A) 72.0 B) 36.1 C) 96.2 D) 144.5
78. In which case Raoult's law is not applicable
- A) 1 M NaCl B) 1 M urea
 C) 1 M glucose D) 1 M sucrose
79. In an experiment, 1 g of a non-volatile solute was dissolved in 100 g of acetone (mol. mass = 58) at 298 K . The vapour pressure of the solution was found to be 192.5 mm Hg . The molecular weight of the solute is (vapour pressure of acetone = 195 mm Hg)
- A) 25.24 B) 35.24
 C) 45.24 D) 55.24
80. A solution has a 1 : 4 mole ratio of pentane to hexane. The vapour pressure of the pure hydrocarbons at 20°C are 440 mm Hg for pentane and 120 mm Hg for hexane. The mole fraction of pentane in the vapour phase would be
- A) 0.549 B) 0.2 C) 0.786 D) 0.478
81. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R
 Assertion A :- Carbon forms two important oxides $-\text{CO}$ and CO_2 . CO is neutral whereas CO_2 is acidic in nature.

Reason R :- CO_2 can combine with water in a limited way to form carbonic acid, while CO is sparingly soluble in water.

In the light of the above statements, choose the most appropriate answer from the options given below :-

- A) Both A and R are correct but R is NOT the correct explanation of A .
 B) Both A and R are correct and R is the correct explanation of A .
 C) A is not correct but R is correct.
 D) A is correct but R is not correct.
82. Choose the correct statement(s) among the following.
 (A) $SnCl_2 \cdot 2H_2O$ is a reducing agent.
 (B) SnO_2 reacts with KOH to form $K_2[Sn(OH)_6]$.
 (C) A solution of $PbCl_2$ in HCl contains Pb^{2+} and Cl^- ions.
 (D) The reaction of Pb_3O_4 with hot dilute nitric acid to give PbO_2 is a redox reaction.
- A) A, C B) A, D C) A, B D) A, B, C
83. Select correct statement (s).
 A) Cyanamide ion (CN_2^{2-}) is isoelectronic with CO_2 and has the same linear structure
 B) Mg_2C_3 reacts with water to form propyne
 C) CaC_2 has $NaCl$ type lattice
 D) All of the above
84. Which of the following compound formed silicones on hydrolysis
 A) $(CH_3)SiCl_2$ B) $(SiH_3)_3N$
 C) $SiCl_4$ D) All
85. Assertion : $Al(OH)_3$ is insoluble in NH_4OH but soluble in $NaOH$.
 Reason : $NaOH$ is strong alkali.
 A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 C) If the Assertion is correct but Reason is incorrect.
 D) If both the Assertion and Reason are incorrect.
86. Heating an aqueous solution of aluminium chloride to dryness will give:
 A) $AlCl_3$ B) Al_2Cl_6
 C) Al_2O_3 D) $Al(OH)Cl_2$

87. The solubility of anhydrous $AlCl_3$ and hydrous $AlCl_3$ in diethyl ether are S_1 and S_2 respectively. Then
 A) $S_1 = S_2$
 B) $S_1 > S_2$
 C) $S_1 < S_2$
 D) $S_1 < S_1$ but not $S_1 = S_2$
88. The species that do not contain peroxide linkage are :-
 A) PbO_2 B) H_2O_2
 C) SrO_2 D) BaO_2
89. Hoop's process is used for the purification of the metal
 A) Al B) Zn C) Ag D) Cu
90. Alumina is
 A) Acidic B) Basic
 C) Amphoteric D) None of these

Biology - (Zoology)

91. Who proposed that the genetic code for amino acids should be made up of three nucleotides?
 A) George Gamow B) Francis Crick
 C) Jacques Monod D) Franklin Stahl
92. Given below are two statements :
 Statement I : Transfer RNAs and ribosomal RNA do not interact with mRNA.
 Statement II : RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.
 In the light of the above statements, choose the most appropriate answer from the options given below :
 A) Both statement I and statement II are correct
 B) Both statement I and statement II are incorrect
 C) Statement I is correct but statement II is incorrect
 D) Statement I is incorrect but statement II is correct

93. Given below are two statements :

Statement I : In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II: DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below :

- A) Both statement I and statement II are correct
- B) Both statement I and statement II are incorrect
- C) Statement I is correct but statement II is incorrect
- D) Statement I is incorrect but statement II is correct

94. Which factor is important for termination of transcription?

- A) α (alpha)
- B) σ (sigma)
- C) ρ (rho)
- D) γ (gamma)

95. Match List I with List II:

List - I	List - II
(a) Alfred Hershey and Martha Chase	(i) Streptococcus pneumoniae
(b) Euchromatin	(ii) Densely packed and dark-stained
(c) Frederick Griffith	(iii) Loosely packed and light-stained
(d) Heterochromatin	(iv) DNA as genetic material confirmation

Choose the correct answer from the options given below:

- A) A-II, B-IV, C-I, D-III
- B) A-IV, B-II, C-I, D-III
- C) A-IV, B-III, C-I, D-II
- D) A-III, B-II, C-IV, D-I

96. Which chromosome in the human genome has the highest number of genes?

- A) Chromosome X
- B) Chromosome Y
- C) Chromosome 1
- D) Chromosome 10

97. Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?

- A) They are monozygotic twins.
- B) They are fraternal twins.
- C) They were conceived through in vitro fertilization.
- D) They have 75% identical genetic content.

98. Histones are enriched with -

- A) Lysine & Arginine
- B) Leucine & Lysine
- C) Phenylalanine & Leucine
- D) Phenylalanine & Arginine

99. Which of the following are the post-transcriptional events in an eukaryotic cell?

- A. Transport of pre-mRNA to cytoplasm prior to splicing.
- B. Removal of introns and joining of exons.
- C. Addition of methyl group at 5' end of hnRNA.
- D. Addition of adenine residues at 3' end of hnRNA.
- E. Base pairing of two complementary RNAs.

Choose the correct answer from the options given below :

- A) A, B, C only
- B) B, C, D only
- C) B, C, E only
- D) C, D, E only

100. Given below are some stages of human evolution.

Arrange them in correct sequence. (Past to Recent)

- A. Homo habilis B. Homo sapiens
 - C. Homo neanderthalensis D. Homo erectus
- Choose the correct sequence of human evolution from the options given below:

- A) B – A – D – C
- B) C – B – D – A
- C) A – D – C – B
- D) D – A – C – B

101. The Permian period, during which the first most modern orders of insects appeared, occurred approximately

- A) 80 million years ago
- B) 150 million years ago
- C) 280 million years ago
- D) 550 million years ago

102. Which of the following is not the result of convergent evolution?

- A) Forelimbs of whales and bats

- B) Flippers of penguins and dolphins
C) Potato and sweet potato
D) Eye of the octopus and mammals
103. Which theory argues that life on earth came from outer space?
A) Theory of panspermia
B) Cosmozoic theory
C) Spore theory
D) All of these
104. What was not used by Millar in his experiment ?
A) Condenser, electrodes, CO_2 , and O_2 .
B) CH_4 , H_2 , H_2O , NH_3 , boiling water.
C) Electrodes, boiling water, CH_4 , NH_3 .
D) Gases (CH_4 , NH_3 , H_2O , H_2), boiling water, condenser.
105. Random unidirectional change in allele frequencies that occurs by chance in all populations and especially in small populations is known as
A) migration
B) natural selection
C) genetic drift
D) mutation.
106. Which of the following was most likely to have been absent in free form in the primordial atmosphere at the time of origin of life
A) O_2
B) CH_4
C) H_2
D) NH_3
107. Contribution in chemical evolution in origin of life
A) Miller
B) Darwin
C) Lamarck
D) Wallace
108. According to one of the most widely accepted theories the earth's atmosphere, before any life had originated on this planet, consisted of a mixture of
A) Water vapours, H_2 , NH_3 and CH_4
B) H_2 , CO_2 , NH_3 and CH_4
C) O_2 , NH_3 , CH_4 and water vapours
D) Ozone, CH_4 , O_2 and water vapours
109. Given below are two statements:
Statement I:
Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance
Statement II:
Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height
In the light of the above statements, choose the correct answer from the options given below:
A) Both Statement I and Statement II are incorrect
B) Statement I is correct but Statement II is incorrect
C) Statement I is incorrect but Statement II is correct
D) Both Statement I and Statement II are correct
110. Mendel conducted experiments for
A) 7 years
B) 6 years
C) 5 years
D) 4 years
111. A son of a colourblind father and normal homozygous mother marries to a daughter of normal father and heterozygous mother than their children will be
A) All colourblind sons
B) All colourblind sons and daughters
C) Some colourblind and some normal daughters
D) Some normal and some colourblind sons
112. Which of the following has been used for genetic researches
A) Pisum
B) Neurospora
C) E. coli
D) All the above
113. Alleles are form of
A) Genes
B) Chromosomes
C) DNA
D) None
114. The colour of flower in pea is always linked with
A) Cotyledon
B) Pod
C) Leaves
D) Seed coat
115. A 1 : 1 phenotypic ratio in a test cross indicates that the
A) Alleles are codominant
B) Dominant phenotype of parent was heterozygous
C) Alleles segregated independently
D) Alleles are dominant
116. Mendelism is related with
A) Heredity in living beings
B) Meiosis during sexual reproduction
C) Mutation in living organisms
D) None of the above
117. The branch of botany dealing with heredity and variation is called
A) Geobotany
B) Sericulture
C) Genetics
D) Evolution

118. Which of the following techniques are used in IVF?

(I.) ZIFT (II), IUT (III) ICSI (IV.) AI (V.) IUI

- A) II and III B) I and III
C) I and II D) II and III

119. Which of the following is not included under barrier methods of birth control?

- A) Vaginal pouch B) Diaphragm
C) Cervical cap D) Implant

120. Periodic abstinence is avoiding sex during

- A) Luteal phase B) Ovulatory phase
C) Menstrual phase D) None of these

121. Identify the correct statements

- i. Birth control pills are likely to cause cardiovascular problem
ii. A woman who substitutes or takes the place of the real mother to nurse to embryo is called surrogate mother
iii. Numerous children have been produced by in vitro fertilisation but with some abnormalities
iv. Woman plays a key role in the continuity of the family and human species
v. Foetal sex determination test should not be banned

- A) I and II
B) II and IV
C) III and V
D) I, II and IV

122. The permissible use of the technique amniocentesis is for

- A) Detecting any genetic abnormality
B) Detecting sex of the unborn foetus
C) Artificial insemination
D) Transfer for embryo into the uterus of a surrogate mother

123. Action of vaginal diaphragm is

- A) Prevent the ova to come in the uterus
B) Prevent the sperm to come in contact with ova
C) Spermicidal
D) Anti-implantational

124. Amniocentesis is the withdrawal of amniotic fluid in

- A) Menopause B) Lactation
C) Gestation D) Pregnancy

125. The success of birth control programmes in controlling population growth is dependent on

- A) Use of contraceptives
B) Tubectomy
C) Vasectomy
D) Acceptability of the above by the people

126. Population of India in early 21st century may be

- A) 105 crore B) 125 crore
C) 95 crore D) 155 crore

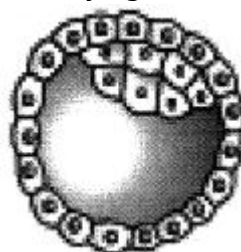
127. Which gland's secretion is act as lubricants in male?

- A) Bulbourethral gland B) Pancrease
C) Seminal vesicle D) Vas deferens

128. Fertilization takes place in

- A) Cervix B) Isthmus C) Ampulla D) Follicle

129. Identify figure



- A) Morula B) Blastocyst
C) Trophoblast D) Blastomere

130. The organ which produces gametes are called ...A... and which neither produces gametes nor hormones are called ...B.... Here A and B represent

- A) A—primary sex organs; B—secondary sex organs
B) A—secondary sex organs; B—primary sex organs
C) A—tertiary sex organs; B—secondary sex organs
D) A—secondary sex organs; B—tertiary sex organs

131. Seminal plasma in human males is rich in

- A) Fructose and calcium
B) Glucose and Calcium
C) DNA and testosterone
D) Ribose and potassium

132. Choose incorrect pair.

- A) Interstitial cells : male sex hormone testosterone
B) Sertoli cells : provide nutrition to the developing sperms
C) Seminal vesicle : produce seminal plasma

- D) Prostate gland : responsible for sperm formation

133. Human uterus is

- A) Paired with well separated oviducts
- B) Single large chamber with posterior part of oviduct fused to it anteriorly
- C) Paired with partially fused oviducts
- D) A single large chamber with completely fused oviducts

134. Testis produce

- A) Sperms B) Eggs C) Seeds D) Spores

135. The rupture of the graafian follicle and the release of ovum occurs under the influence of

- A) LH B) FSH C) MSH D) GH

Biology - (Botany)

136. Which of the following *RNAs* is not required for the synthesis of protein?

- A) mRNA B) tRNA C) rRNA D) siRNA

137. DNA polymorphism forms the basis of :

- A) DNA finger printing
- B) Both genetic mapping and DNA finger printing
- C) Translation
- D) Genetic mapping

138. Which one of the following pair is a purine pair?

- A) Uracil, Guanine
- B) Cytosine, Thymine
- C) Adenine, Guanine
- D) Adenine, Thymine

139. Which of the following is correct for Watson and Crick's model of DNA. It is duplex with

- A) 10 base pairs and 34 Å distance for every turn.
- B) 10 base pairs and 3.4 Å distance for each turn of spiral.
- C) 20 base pairs and 34 Å for each turn.
- D) None of the above

140. Assertion : In a DNA molecule, A – T rich parts melt before G – C rich parts.

Reason : In between A and T there are three H–bond, whereas in between G and C there are two H– bonds.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the

Assertion.

- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

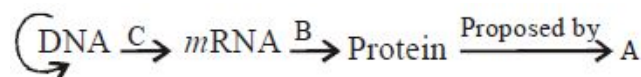
141. Purines found both in DNA and RNA are

- A) Adenine and thymine
- B) Adenine and guanine
- C) Guanine and cytosine
- D) Cytosine and thymine

142. Under which of the following conditions will there be no change in the reading frame of following mRNA 5' AACAGCGGUGCUAUU3'

- A) Insertion of G at 5th position
- B) Deletion of G from 5th position
- C) Insertion of A at G at 4th and 5th positions respectively
- D) Deletion of GGU from 7th, 8th and 9th positions

143. The figure gives an important concept in the genetic implication of DNA. Fill the blanks A, B and C.



- A) A-Maurice Wilkins, B-Transcription, C-- Translation
- B) A-James Watson, B-Replication, C-Extension
- C) A-ErwinChargaff, B-Translation, C-Replication
- D) A-FrancisCrick, B-Translation, C-Transcription

144. Octamer which form complex unit with DNA is called

- A) Nucleosome
- B) Centrosome
- C) Chromosome
- D) Endosome

145. Genetically active area of chromosome is called

- A) Euchromatin
- B) Heterochroatin
- C) Heptan
- D) Cistron

146. The modern concept of gene is

- A) A segment of DNA capable of crossing over
- B) A functional unit of DNA
- C) A segment of DNA
- D) A segment of chromosome

147. Who used heavy nitrogen N^{15} for proving semiconservative way of DNA replication

- A) Masters and Broda
- B) Meselson and Stahl
- C) Watson and Crick
- D) Jacob and Monad

148. Adenine is 30% then what would be the % of guanine

- A) 10% B) 20% C) 30% D) 40%

149. A nucleoside differs from a nucleotide in not having

- A) Phosphate B) Sugar
C) Nitrogen base D) Phosphate and sugar

150. Wild type E.coli cells are growing in normal medium with glucose. They are transferred to a medium containing only lactose as the sugar. Which one of the following changes take place

- A) The lac-Operon is repressed
B) All Operons are induced
C) E. coli cells stop dividing
D) The lac-Operon is induced

151. 'Central Dogma' was proposed by

- A) Crick B) Beadle and Tatum
C) Temin and Baltimore D) Klug

152. Transcription is

- A) Formation of *RNA* from *DNA*
B) Formation of *DNA* from *RNA*
C) Formation of protein
D) Transfer of *RNA*

153. Which of the following is nongenetic, which is utilised for protein synthesis

- A) *DNA* B) *zRNA*
C) *mRNA* D) None of these

154. Which one of the following is common to both prokaryotes and eucaryotes

- A) Mitotic apparatus B) Histones
C) Mitochondria D) Genetic code

155. What is meant by sense strand

- A) The strand of *mRNA* involved in protein synthesis
B) The strand of *tRNA* which starts protein synthesis
C) The *DNA* strand which acts as a template for *mRNA* transcription
D) Gradually growing molecule of protein

156. Isolation and purification of specific *DNA* segment from a living organism was achieved by

- A) Crick B) Nirenberg

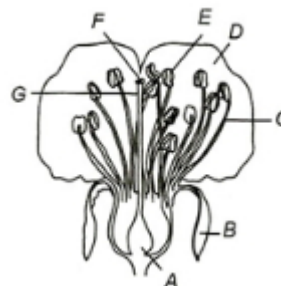
C) Khorana

D) Beckwith and his colleagues

157. The strongest evidence that *DNA* is genetic material comes from

- A) That chromosomes are made up of *DNA*
B) Transformation of bacterial cells
C) That *DNA* is not present in the cytoplasm
D) That *DNA* is present in the nucleus

158. Identify *A* to *G* in following figure and answer accordingly



- A) *A*—Ovary, *B*—Filament, *C*—Sepal, *D*—Petal, *E*—Style, *F*—Stigma, *G*—Anther
B) *A*—Petal, *B*—Ovary, *C*—Petal, *D*—Filament, *E*—Anther, *F*—Stigma, *G*—Style
C) *A*—Ovary, *B*—Sepal, *C*—Filament, *D*—Petal, *E*—Anther, *F*—Stigma, *G*—Style
D) *A*—Ovary, *B*—Sepal, *C*—Filament, *D*—Petal, *E*—Anther, *F*—Stigma, *G*—Style

159. Which of the following is not a characteristic feature of insect pollinated flowers?

- A) Fragrance
B) Nectaries
C) Foul odour
D) Mucilaginous covering on pollen grains

160. The process of formation of microspores from a pollen mother cell is called

- A) Megasporogenesis B) Microsporogenesis
C) Megagametogenesis D) Microgametogenesis

161. Which one of the following would not lead to formation of clones?

- A) Double fertilization B) Apomixis
C) Vegetative reproduction D) Tissue culture

162. Sexual reproduction leads to

- A) Genetic recombination B) Polyploidy
C) Aneuploidy D) euploidy

163. Flower is a

- A) Modified male plant only
B) Modified female plant only

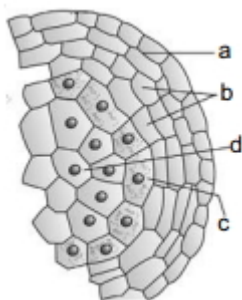
- C) Modified reproductive shoot
- D) Vegetative shoot system

164. In flowering plants meiosis occurs at the time of

- A) Formation of buds
- B) Germination of seed
- C) Formation of root primordia
- D) Formation of pollen grains

165. Examine the figure given below and select the right option giving all the four parts *a, b, c* and *d*. Correctly identify

a b c d



- A) Endothecium Tapetum Microspore mother cell Middle layers
- B) Tapetum Endothecium Microspore mother cell Middle layers
- C) Endothecium Middle layer Tapetum Microspore mother cell
- D) Endothecium Microspore mother cell Middle layer Tapetum

166. Which of the following statement is applicable for all flowering plants?

- A) Monosiphonous pollen tube
- B) Non-motile and morphologically dissimilar gametes
- C) Presence of pollinium
- D) Division of generative cell after pollination

167. Seed germination requires

I. Light II. Temp (suitable)
III. Moisture IV. Oxygen
Select correct option

- A) *I, II and III*
- B) *II, III and IV*
- C) *I, III and IV*
- D) *II, IV and I*

168. What does the filiform apparatus do at the entrance into ovule?

- A) It brings about opening of the pollen tube.
- B) It guides pollen tube from a synergid to egg.
- C) It helps in the entry of pollen tube into a synergid.

- D) It prevents entry of more than one pollen tube into the embryo sac.

169. Diameter of pollen grain

- A) 25 – 50 $\text{\AA}$
- B) 25 – 50 micron
- C) 20 – 25 micrometre
- D) 25 – 50 micrometre

170. Assertion : Endosperm is a nutritive tissue and it is triploid.

Reason: Endosperm is formed by fusion of secondary nucleus to second male gamete. It is used by developing embryo

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

171. A- Four cells are at the chalzal end and are called the antipodals.

R – A typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled

- A) A – R correct
- B) A – R incorrect
- C) A -correct, R -incorrect
- D) A -incorrect, R -correct

172. Because of it pollen grains are well preserved as fossils

- A) Presence of germ pore
- B) Presence of sporopollenin
- C) Presence of cellulose
- D) All correct

173. Parthenocarpic tomato fruits can be produced by

- A) treating the plants with phenylmercuric acetate
- B) removing androecium of flowers before pollen grains are released
- C) treating the plants with low concentrations of gibberellic acid and auxins
- D) raising the plants from vernalized seeds.

174. Which of the following statements is correct?

- A) Sporopollenin can be degraded by enzymes.
- B) Sporopollenin is made up of inorganic materials.

